

YILU WANG

M.Sc. Candidate, Molecular Biosciences (Cancer Biology) · DKFZ & Heidelberg University
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EDUCATION

Oct 2025 – Present **Heidelberg University & German Cancer Research Center (DKFZ)**

M.Sc. in Molecular Biosciences · Major: Cancer Biology

- Grade average 1.3 / 1.0 (top band “very good”).
- Hans-Peter Wild Talent Scholarship (2025).

Sep 2021 – Jun 2025 **Tsinghua University, Beijing, China**

B.S. in Biochemistry & Biomedical Engineering (double major)

- Grade average 3.7 / 4.0 (top 10%).
- University excellence scholarships (2022 – 2024).

RESEARCH EXPERIENCE

Jun 2026 – Present **Institute for Computational Biomedicine & TSPC, Heidelberg University Hospital**

Research Intern · Advisor: Prof. Denis Schapiro (Spatial omics)

- Comparison and optimization of data-integration methods for imaging-based spatial proteomics.

Nov 2024 – Feb 2025 **Harvard University — School of Engineering & Applied Sciences**

Research Intern · Advisor: Prof. David Weitz (Experimental soft condensed matter)

- Developed hydrogel-bead-based in vitro display systems for nanobody discovery; constructed and screened a $\sim 10^{13}$ -variant library by high-throughput FACS, recovering binders with ultra-high affinity and specificity.

Jul – Aug 2024 **Stanford University — Department of Mechanical Engineering**

Research Intern · Advisor: Prof. Juan Santiago (Microfluidics)

- Built tube-embedded microfluidic devices for electrokinetic cell-free DNA purification (isotachophoresis) with TaqMan qPCR readout; imaged single-molecule DNA dynamics in PDMS chips.

Sep 2022 – Jun 2024 **Tsinghua University — Institute of Medical Systems Biology**

Undergraduate Researcher · Advisor: Prof. Yong Guo (Molecular diagnostics)

- Developed multi-target digital PCR assays for robust HER2 detection in FFPE and blood samples, with a Python watershed pipeline for droplet clustering.

PUBLICATIONS

- [1] Q. Jiang, X. Zang, **Y. Wang**, A. S. Avaro, D. A. Huyke, J. G. Santiago. “A three-dimensional microfluidic device embedded within a thermal cycler tube for electrokinetic DNA extraction.” *Lab on a Chip*, 2025.
- [2] **Y. Wang**, Y. Guo. “Single-tube single-, dual-, and triple-target digital PCR assays for detection of HER2 amplification.” B.S. Thesis, Tsinghua University, 2025.
- [3] X. Chen, et al., **Y. Wang**, Y. Guo, Z. Xie. “Development and evaluation of a multiplex digital PCR method for sensitive and accurate detection of respiratory pathogens in children.” *Virology*, 2024.
- [4] H. Yang, et al., **Y. Wang**, G. Huang. “Graphene-based flexible biosensing technology and wearable precision medical-health-monitoring application.” *Chinese Journal of Lasers*, 2024.

SCHOLARSHIPS & AWARDS

2025	Hans-Peter Wild Talent Scholarship, Heidelberg University
2022 – 2024	Comprehensive Excellence Scholarship, Tsinghua University (top 2% of undergraduates)
2023	Academic Research Excellence Scholarship, Tsinghua University (top 1% of undergraduates)
2023	Gold Medal & nominations for Best Therapeutics Project / Best Software Tool, iGEM Competition
2024	Second Prize, Tsinghua 42nd Challenge Cup (Student Academic & Scientific Works)
2023	Independent Investigator, Qiyan Program Fund, Beijing Natural Science Foundation
2020	National high-school Olympiads: First Prize in Math & Physics (Heilongjiang); Silver Medal, China Girls' Mathematics Olympiad

EXTRACURRICULAR ACTIVITIES

2022 – 2025	Qingxin Times, Tsinghua University <i>Chief Reporter, Director & Consultant</i> · Named Most Influential Student Journalist (2022); awarded University Journalism Awards for Best Writing (2023); wrote, edited and published 20+ feature articles across campus and external press.
2023	Rural Revitalization Workstation, Tsinghua University <i>Team Leader</i> · Led two field teams researching and supporting rural communities in Shaanxi, China; awarded Best Student Social Practice (2022, 2023).
2023 – 2024	International Genetically Engineered Machine (iGEM) <i>Team Leader (2023) & Advisor (2024)</i> · Led and mentored synthetic-biology therapeutics teams.

PROFESSIONAL SKILLS

Experimental	Molecular & cell biology (PCR / qPCR / digital PCR, cloning, gel electrophoresis, western blot, ELISA, etc.); cell culture (cancer lines, T cells, iPSCs, neurons, etc.) & engineering (AAV, retrovirus, lentivirus); flow cytometry / FACS; microscopy (fluorescence, confocal); microfluidics (PDMS chip design & fabrication, droplet sorting).
Computational	Multi-omics data analysis — whole genome- / exome-seq, RNA-seq, ATAC-seq, gene-enrichment analysis, spatial transcriptomics / proteomics; machine learning / neural networks. Programming: Python, R, C/C++, MATLAB.
Languages	Chinese (native) · English (fluent) · Japanese (intermediate) · German (elementary)

GRADUATE LABORATORY TRAINING

Winter 2025	Cancer immunology, stem-cell biology & functional genomics <i>Practical course (HP-F2), DKFZ</i> · Three lab rotations: engineering by AAV delivery and cytotoxicity testing of CAR-T cells (P. Schmidt); characterization of inflammatory-stress response of hematopoietic stem cells involving retrovirus delivery and flow cytometry (M. Essers); and RNA-seq differential-expression analysis in R with DESeq2 (C. Erkut).
Summer 2026	Cell reprogramming, translation & RNA modification <i>Practical course (HP-F7), DKFZ</i>

- Four lab rotations: reprogramming induced pluripotent stem cells (iPSCs) into neurons by lentivirus followed by chromatin profiling (ATAC-seq) (M. Mall & D. Odom); ribosome profiling of protein translation (F. Loayza-Puch); and characterization of inhibitor-induced m⁶A RNA-methylation in melanoma involving immunofluorescence microscopy (M. Frye).